

PORT ANGELES FAIRCHILD, WA, USA

WMO: 727885

Lat: 48.120N

Lon: 123.498W

Elev: 288

StdP: 14.54

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94266

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.3	27.8	15.4	12.1	30.3	19.8	15.1	31.2	20.0	44.1	17.3	43.7	4.2	230	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.3	78.8	62.0	73.4	60.2	70.1	59.3	63.2	74.5	61.7	71.2	60.3	68.2	5.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	75.5	64.9	57.4	71.0	63.5	56.6	69.0	63.0	28.7	74.3	27.5	71.2	26.6	68.2	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.4	15.1	12.6	DB	20.2	87.3	4.2	4.1	17.2	90.3	14.7	92.7	12.4	95.0	9.4	98.0
			WB	18.7	66.7	4.1	2.4	15.8	68.5	13.4	69.9	11.1	71.3	8.2	73.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	39.9	40.1	42.8	46.7	51.8	56.0	59.9	60.4	57.1	49.3	43.2	39.3
	DBStd	8.81	4.91	4.74	4.16	3.94	4.32	4.37	3.88	3.58	3.88	4.19	5.15	4.68
	HDD50	1571	315	278	227	113	30	2	0	0	1	62	211	332
	HDD65	5901	779	697	690	549	410	277	168	151	241	487	655	797
	CDD50	1172	1	1	2	14	85	181	307	323	212	40	6	0
	CDD65	26	0	0	0	0	0	5	10	9	2	0	0	0
	CDH74	382	0	0	0	0	6	72	131	137	36	0	0	0
	CDH80	92	0	0	0	0	0	16	37	34	5	0	0	0
Wind	WSAvg	5.3	4.4	4.5	5.1	6.0	6.2	6.5	6.6	5.7	4.6	4.2	4.4	4.7
Precipitation	PrecAvg	32.8	5.3	3.1	3.0	1.9	1.4	1.0	0.6	1.0	0.8	3.5	6.5	5.1
	PrecMax	45.4	8.7	8.7	6.7	4.1	2.9	2.4	1.5	2.6	2.3	8.7	14.2	7.8
	PrecMin	20.6	2.0	0.0	0.9	0.5	0.2	0.3	0.2	0.2	0.0	0.8	1.6	2.6
	PrecStd	6.6	2.0	1.9	1.4	0.9	0.7	0.4	0.4	0.8	0.6	1.9	3.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	55.4	59.3	67.3	73.8	81.9	84.6	84.3	78.5	65.3	57.5	53.5
		MCWB	49.0	48.1	49.7	53.9	56.8	60.7	64.7	63.2	61.5	54.7	53.6	50.3
	2%	DB	50.8	51.6	54.7	61.2	68.0	73.5	77.2	78.0	72.9	61.4	54.7	49.9
		MCWB	47.8	46.8	48.3	50.6	55.5	59.5	62.1	62.0	59.4	53.2	51.5	47.1
	5%	DB	48.4	49.2	52.2	57.1	63.6	67.6	71.9	72.5	69.5	58.7	52.4	47.6
		MCWB	46.0	45.7	46.8	49.3	54.4	57.6	60.4	60.9	58.6	52.8	49.9	45.2
	10%	DB	46.5	47.4	50.0	54.3	60.1	63.6	68.2	68.8	65.7	56.5	50.4	45.9
		MCWB	44.4	44.3	45.7	48.0	53.0	55.9	59.3	59.7	57.6	51.9	48.0	43.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	50.3	51.8	54.8	59.2	62.9	66.4	65.0	63.8	58.8	55.1	50.9
		MCDB	52.3	54.1	56.3	64.5	67.9	75.9	81.6	77.8	73.7	62.9	56.5	52.0
	2%	WB	48.3	48.0	49.4	52.0	57.1	60.3	63.2	63.1	61.3	55.8	52.6	47.3
		MCDB	49.8	50.6	53.1	58.5	65.4	71.2	74.7	74.5	69.6	58.7	53.9	48.9
	5%	WB	46.4	46.3	47.8	50.4	55.1	58.5	61.2	61.7	59.8	54.0	50.4	45.8
		MCDB	47.8	48.5	51.2	55.8	62.2	66.8	70.4	71.0	66.6	56.7	51.9	47.0
	10%	WB	45.2	44.8	46.3	48.7	53.4	56.7	59.7	60.3	58.4	52.6	48.4	44.3
		MCDB	46.6	46.9	49.3	53.4	59.0	63.1	67.1	67.8	64.2	55.5	50.1	45.8
Mean Daily Temperature Range	5% DB	MDBR	10.3	11.7	12.7	13.8	13.9	14.1	15.7	16.3	16.4	13.4	11.5	9.9
		MCDBR	11.8	13.0	15.4	19.2	21.7	23.5	24.5	25.2	23.7	16.5	11.8	10.7
		MCWBR	9.0	9.3	10.2	11.2	11.9	11.5	11.9	12.1	12.5	10.3	9.0	8.5
	5% WB	MCDBR	10.4	11.4	13.8	16.8	17.8	19.2	20.2	19.8	18.7	13.2	10.6	9.7
		MCWBR	8.3	8.6	9.8	10.4	10.4	10.2	10.4	10.3	10.8	9.5	8.7	8.3
Clear-Sky Solar Irradiance	taub	0.318	0.319	0.332	0.360	0.383	0.361	0.351	0.354	0.341	0.333	0.330	0.316	
	taud	2.494	2.518	2.497	2.415	2.370	2.454	2.518	2.510	2.546	2.552	2.471	2.453	
	Ebn at Noon	241	269	282	283	279	285	287	282	276	260	232	224	
	Edh at Noon	20	24	29	34	37	34	32	31	27	23	20	18	
All-Sky Solar Radiation	RadAvg	303	581	895	1339	1706	1865	2052	1739	1249	692	359	245	
	RadStd	41	78	105	125	136	182	148	114	107	75	52	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-1.44	-4.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page